Architectural Isolation as a Timing Safety Primitive for Edge AI Medical Devices: Controlled Experimental Evidence on a Shared-Silicon Platform

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***Abstract*—Output correctness and timing reliability are independent safety properties of edge AI inference pipelines, yet FDA pre-market validation does not explicitly characterize either under deployment conditions. This letter demonstrates their independence through a controlled same-hardware experiment: identical MobileNetV2 models are evaluated under identical adversarial load on two execution paths of the same NVIDIA Jetson Orin Nano Super: a dedicated GPU accelerator (TensorRT FP16) and a general-purpose CPU (ONNX Runtime FP32). Across five stressor classes and $\approx$158,000 inference activations, both paths maintain $\mathrm{STER}{=}0.0000$ (zero exceedances above $T^{*}{=}0.05$), confirming output instability is not the dominant failure mode. The paths diverge sharply on timing: the GPU path maintains latency <15 ms under all conditions; the CPU path degrades $7.2\times$ under combined load (14.5 ms to 104.0 ms mean, $\mathrm{P99}{=}165.1$ ms), breaching the 10 Hz clinical cycle budget by 65%. Because the same-hardware design eliminates chip-to-chip confounds, this divergence is consistent with causal attribution to the dedicated accelerator architecture. The Safety-Threshold Exceedance Rate (STER), formally defined as the proportion of inferences exceeding a clinically-motivated $\pm5\%$ output deviation tolerance, provides a mechanism to verify output stability independently of timing, enabling isolation of timing failures from functional correctness failures. Joint STER + latency verification is proposed as a candidate pre-market acceptance protocol under FDA Draft Guidance FDA-2024-D-4488 (January 2025).**

***Index Terms*—edge AI, inference stability, timing safety, architectural isolation, Safety-Threshold Exceedance Rate (STER), Software as a Medical Device (SaMD), pre-market validation, IEC 62304, ISO 14971, TensorRT, ONNX Runtime.**

I. Introduction

FDA pre-market validation of AI-enabled medical devices evaluates model accuracy (sensitivity, specificity, AUC) under controlled, zero-contention conditions [1]. This framework does not explicitly characterize whether inference outputs remain within bounds, or whether inference timing is preserved within clinical cycle budgets, when the device operates under realistic deployment load. A 2025 cross-sectional study of 950 FDA-cleared AI-enabled devices found that diagnostic errors were the leading cause of AI device recalls, with 43% occurring within the first year of clearance [15]. IEC 62304 Clause 6.3.2 requires verification that software meets its timing requirements under all foreseeable operating conditions [8]; however, no standardized empirical method exists for characterizing whether inference pipeline timing is preserved under realistic deployment stress.

This letter addresses this gap through a controlled experiment. Running identical models under identical stressors on both the GPU accelerator and the general-purpose CPU of the same physical hardware eliminates platform confounds and supports attribution of timing behavior to the accelerator architecture. This work makes four contributions:

1) **Output/timing independence**: output correctness (STER = 0.0000) is preserved on both paths under all conditions; timing reliability diverges sharply. Current accuracy-based validation may not detect this distinction ($\Delta\mathrm{acc}{<}1.0\%$ on both paths).
2) **Controlled same-hardware evidence**: GPU path maintains STER = 0.0000 and latency <15 ms; CPU path maintains STER = 0.0000 but degrades to 104 ms mean / 165 ms P99 under combined load, breaching the 10 Hz clinical cycle budget by 65%.
3) **STER as output stability verification**: STER formally verifies that inference output distributions remain within a clinically-motivated tolerance band, independently of timing, enabling isolation of timing failures from correctness failures.
4) **Joint pre-market protocol**: $\mathrm{STER} \leq \mathrm{STER}_{\max}$ AND $\mathrm{latency}_{\mathrm{P99}} \leq T_n$ under foreseeable stressor conditions is proposed as a reproducible acceptance criterion operationalizing FDA-2024-D-4488's robustness assessment requirement.

II. Related Work

Prior edge inference work addresses throughput maximization [3], latency predictability under GPU partitioning [4], and cross-device accuracy benchmarking [5], [6]. These works measure latency *or* output accuracy, not their independence under shared-silicon contention. Safety-oriented work on neural networks in medical devices [7] addresses model-level mechanisms but does not empirically characterize resource-contention-induced timing failure as a distinct safety property from output correctness. No prior study has used a same-hardware controlled design to isolate the architectural contribution of a dedicated accelerator to timing safety [4].

III. System Model and Formal Definitions

A. Output Stability Under Identical Inputs

Let f denote a deployed inference model and $\mathbf{x}_i$ a fixed test input ($i = 1, \ldots, N$). Under a stressor condition c,

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let $\sigma(\mathbf{y}_i^c)$ denote the softmax output and $\bar{\sigma}_i$ the per-image reference softmax captured at zero load. The per-inference output deviation is:

$$\delta_i = \left\|\sigma(\mathbf{y}_i^c) - \bar{\sigma}_i\right\|_\infty = \max_k \left|\sigma(\mathbf{y}_i^c)_k - \bar{\sigma}_{i,k}\right| \quad (1)$$

B. STER Definition

Definition 1 (T^*): $T^* = 0.05$, selected as conservative relative to ISO 14971 Table B.1 clinical diagnostic tolerance bands (typically 10–20%).

Definition 2 (STER): For N inference activations:

$$\mathrm{STER} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}[\delta_i > T^*] \quad (2)$$

STER verifies output distribution bounds, which accuracy ($\mathrm{argmax}(\sigma(\mathbf{y}))$, a single label) structurally cannot confirm. STER is proposed as a candidate pre-market metric, not an existing FDA-specified standard.

C. Robustness Confirmation Condition

The joint pre-market acceptance condition is:

$$\mathrm{STER}_c \leq \mathrm{STER}_{\max} \quad \text{AND} \quad \mathrm{latency}_{\mathrm{P99}} \leq T_n \quad (3)$$

where $\mathrm{STER}_{\max} = 0$ and $T_n = 100\,\mathrm{ms}$ (10 Hz cycle budget). STER and latency are orthogonal safety dimensions: a device may satisfy (3) fully (GPU path, E0–E5), satisfy only the STER term (CPU path, E6), or fail both. Both must be verified.

D. Architectural Model

The Jetson Orin Nano Super uses shared LPDDR5: CPU, GPU, and memory controller compete for the same bus. The *GPU execution path* (TensorRT FP16) offloads inference to the 1024-core Ampere GPU via dedicated DMA paths with QoS bandwidth reservation, isolating inference from CPU scheduling pressure. The *CPU execution path* (ONNX Runtime FP32) runs on the 6-core Arm Cortex-A78AE cluster, directly competing with stressor workloads for CPU time, cache, and DRAM bandwidth. This architectural difference is hypothesized to produce divergent timing behavior under contention while preserving output correctness on both paths; this is empirically tested in E1–E6.

IV. Hardware Platform and Experimental Protocol

A. Hardware Platform

Both execution paths run MobileNetV2 [12] (224×224 RGB, ImageNet 1000-class head) on the same 500-image fixed test set, held constant across all experiments. The GPU path runs TensorRT FP16 [13] at a nominal 10 Hz ($T_n = 100\,\mathrm{ms}$); the CPU path runs MobileNetV2 FP32 via ONNX Runtime [14] on the same hardware. Using the same physical device for both paths is the critical methodological choice: it eliminates chip-to-chip variation and thermal differences as confounds, supporting attribution of timing behavior to the accelerator architecture.

TABLE I
Hardware Platform Summary

Component	Role	Key Specification
Jetson Orin Nano Super 8GB (GPU path)	Primary inference host	1024-core Ampere GPU, TensorRT FP16, shared LPDDR5
Jetson Orin Nano Super 8GB (CPU path)	Negative control (same hardware)	ONNX Runtime FP32, CPUExecutionProvider, Cortex-A78AE
nRF52840 DK + TI SensorTag	BLE network stressor	BLE 5.0, up to 6 simultaneous connections
GL-AX1800 WiFi 6 Router	RF environment	2.4 GHz co-channel interference

TABLE II
Experiment Summary

ID	Stressor	Protocol	IEC
E0	Baseline	Both paths, zero load. Per-image reference $\bar{\sigma}_i$ captured. 10 trials × 500 inferences.	B+C
E1	CPU stress	stress-ng at 25/50/75/100% system-wide load. Both paths tested at each level. 10 trials × 500 inferences per level.	B
E2–E4	Isolation stressors	Memory pressure (25–90% fill), GPU co-tenancy (0–4 workers), network I/O (BLE 0–6 conns + WiFi). GPU path only; STER = 0.0000 across all conditions. 5–10 trials per level.	B/C
E5	Combined load	CPU 75% + memory 50% + BLE 4 conns + fio disk write. Both paths. 10 trials × 500 inferences. *Worst-case deployment scenario.*	C
E6	CPU negative control	CPU path only (no GPU). Conditions matching E0 + E1 + E5. 10 trials × 500 inferences per condition. Primary contrastive result.	B

B. Experimental Protocol

For each trial, $N = 500$ inference activations are recorded. For each activation i: (1) fixed test input $\mathbf{x}_i$ is presented; (2) softmax $\sigma(\mathbf{y}_i)$ is recorded; (3) δ_i is computed per (1) against the per-image zero-load reference; (4) latency is recorded via `time.perf_counter()`. STER is computed per (2) over the N activations. Top-1 accuracy is computed as a control variable.

V. Results

A. E0: Baseline Calibration

Both paths establish matched zero-load baselines (Δlatency <0.3 ms), confirming controlled experimental conditions. GPU floating-point variance produces δ_{mean}=0.0181 but zero exceedances above T^*=0.05; CPU ONNX Runtime is arithmetically deterministic (δ=0.0000 exactly). Top-1 accuracy: 1.6%

TABLE III
E0 BASELINE: BOTH EXECUTION PATHS

Path	STER	δ_{mean}	Lat. mean	Lat. P99
GPU (TensorRT FP16)	0.0000	0.0181	14.8 ms	≈18 ms
CPU (ONNX FP32)	0.0000	0.0000	14.5 ms	18.6 ms

TABLE IV
E1 CPU STRESS: BOTH PATHS (10 TRIALS × 500 INFERENCES PER LEVEL)

CPU Load	GPU Path		CPU Path	
	STER	Lat. mean	STER	Lat. mean
25%	0.0000	≈14.8 ms	0.0000	33.4 ms
50%	0.0000	≈14.8 ms	0.0000	57.8 ms
75%	0.0000	≈14.8 ms	0.0000	73.7 ms
100%	0.0000	≈14.8 ms	0.0000	70.9 ms

on both paths (same model, same images, same label mapping; accuracy values reflect ImageNet label remapping and are not used as performance benchmarks).

B. E1: CPU Stress and Timing Divergence

Both paths maintain STER = 0.0000 at all CPU load levels; δ_{mean} is load-invariant on both paths. Timing diverges immediately: the GPU path is invariant (≈14.8 ms regardless of load) while the CPU path scales with contention, reaching 73.7 ms at 75% load, already $5.1\times$ the zero-load baseline. TensorRT FP16 executes entirely on the GPU; the 6-core Arm cores saturated by stress-ng have no architectural path to disturb the inference pipeline. ONNX Runtime FP32 competes directly with the stressor for CPU time slices and cache bandwidth, producing latency degradation without affecting arithmetic output.

C. E2–E4: Isolation Stressors (GPU Path)

Memory pressure (25–90% LPDDR5 fill), GPU co-tenancy (0–4 concurrent TensorRT workers), and network I/O (BLE 0–6 connections with WiFi co-channel interference) all produce STER = 0.0000 with δ_{mean} indistinguishable from E0 baseline on the GPU path (δ_{mean}=0.0181 ± 0.0002 across all conditions, n>25,000 inferences per stressor class). GPU DMA isolation, dedicated QoS bandwidth reservation, and independent TensorRT execution contexts prevent all tested stressor classes from perturbing inference outputs or latency. A confirmed +88% CPU interrupt rate increase at BLE conns = 4 (389 → 732 IRQ/min) produced no measurable effect on GPU path timing.

D. E5: Combined Realistic Load

Under simultaneous CPU 75% + memory 50% fill + BLE 4 connections + fio disk write, the GPU path maintains STER = 0.0000 and latency <15 ms (mean 14.9 ms, δ_{mean}=0.0156 ± 0.0001) across all 10 trials. The CPU path maintains STER = 0.0000 but reaches 104.0 ms mean / 165.1 ms P99, exceeding the 10 Hz cycle budget by 65%.

TABLE V
E6 CPU-ONLY NEGATIVE CONTROL: JETSON ORIN NANO SUPER, ONNX RUNTIME FP32 (10 TRIALS × 500 INFERENCES PER CONDITION)

Condition	STER	δ_{mean}	Lat. mean	Lat. P99
Zero load	0.0000	0.0000	14.5 ms	18.6 ms
CPU 25%	0.0000	0.0000	33.4 ms	111.4 ms
CPU 50%	0.0000	0.0000	57.8 ms	115.8 ms
CPU 75%	0.0000	0.0000	73.7 ms	122.1 ms
CPU 100%	0.0000	0.0000	70.9 ms	117.0 ms
Combined	**0.0000**	**0.0000**	**104.0 ms**	**165.1 ms**†

† Exceeds T_n=100 ms cycle budget by 65%.

TABLE VI
SUMMARY: GPU VS. CPU PATH UNDER WORST-CASE CONDITIONS

Path / Condition	STER	δ_{mean}	Lat. mean	Lat. P99
GPU E0 (baseline)	0.0000	0.0181	14.8 ms	≈18 ms
GPU E5 (combined)	0.0000	0.0156	14.9 ms	≈18 ms
CPU E0 (baseline)	0.0000	0.0000	14.5 ms	18.6 ms
CPU E5/E6 (combined)	0.0000	0.0000	104.0 ms	165.1 ms†

† Exceeds T_n by 65%. STER passes; deployment timing fails.

E. E6: CPU Negative Control

STER = 0.0000 and δ=0.0000 exactly across all 50,000 inferences: ONNX Runtime's CPUExecutionProvider uses deterministic ARM NEON SIMD operations; scheduling pressure delays computation without perturbing its arithmetic. Latency degrades monotonically with stressor intensity, reaching $7.2\times$ baseline under combined load. P99 = 165.1 ms under combined load means 1% of inferences exceed the clinical cycle budget by ≥65%. The GPU path, under identical combined load, maintains latency <15 ms across all 10 trials. E6 establishes that the dedicated GPU accelerator's contribution to the safety profile is *timing isolation*, not numerical stability; this is a property of deterministic floating-point execution regardless of accelerator.

F. Summary

VI. DISCUSSION

A. Controlled Design as Methodological Contribution

The same-hardware design is the critical methodological choice. Cross-platform comparisons (e.g., Jetson vs. a different board) cannot establish causality because thermal, memory topology, and driver-stack differences confound the result. Running GPU and CPU paths on the same Jetson Orin Nano Super controls for all hardware-level confounds; the timing divergence of $7.2\times$ under identical stressors is consistent with causal attribution to the accelerator architecture. Platform datasheets document architectural separation in principle; this work provides empirical quantification across simultaneous multi-class stressor combinations at clinical deployment intensities.

B. Timing as the Dominant Failure Mode

STER = 0.0000 on both paths establishes that output instability is not the failure mode under realistic resource contention on

this hardware. The dominant failure mode is timing degradation, which is invisible to STER, invisible to accuracy, and invisible to any output-only evaluation method. This result motivates explicit timing characterization as a complementary pre-market verification property, not as a replacement for output stability metrics.

C. Pre-Market Validation Gap

Standard accuracy-based FDA pre-market testing would pass both execution paths without distinguishing their timing safety profiles: $\Delta\mathrm{acc}<1.0\%$ under all conditions on both paths. A manufacturer using accuracy-only testing would clear both a GPU-accelerated and a CPU-only deployment as equivalent, but the CPU-only deployment breaches the 10 Hz clinical cycle budget by 65% under combined load. The joint acceptance condition (3) closes this gap with reproducible, hardware-agnostic criteria.

D. STER as Output Stability Certification

STER certifies that the full output distribution $\sigma(\mathbf{y})$ remains within the T^* tolerance band, a property that $\mathrm{argmax}(\sigma(\mathbf{y}))$ (accuracy) structurally cannot confirm. A model whose top-1 prediction is unchanged but whose confidence distribution shifts by 4.9% passes accuracy testing but may violate T^*; STER detects this; accuracy does not. STER = 0.0000 on both paths in this study confirms output stability under all tested conditions and isolates timing as the sole safety-relevant divergence.

E. Limitations

Results are specific to MobileNetV2 at 10 Hz on Jetson Orin Nano Super under JetPack 6. Models with higher memory bandwidth utilization, non-deterministic preprocessing, or non-linear latency-throughput profiles may exhibit different behavior. $T^*=0.05$ is conservative relative to ISO 14971 Table B.1 but is not clinically validated for any specific device class; the appropriate T^* is a manufacturer and clinical risk management decision.

VII. Conclusion

This letter has demonstrated, through a controlled same-hardware experiment, that output correctness and timing reliability are independent safety properties of edge AI inference pipelines. Running MobileNetV2 under identical adversarial stressors on the GPU and CPU execution paths of the same Jetson Orin Nano Super, both paths maintain STER = 0.0000 across $\approx 158{,}000$ inference activations; the GPU path maintains latency <15 ms while the CPU path degrades $7.2\times$ to 104.0 ms mean / 165.1 ms P99 under combined load, breaching the 10 Hz clinical cycle budget. Current accuracy-based FDA pre-market testing would pass both paths without detecting this timing failure. STER formally verifies output distribution bounds; latency verification under contention is required for timing safety. These results indicate that ensuring output correctness alone may be insufficient for reliable clinical deployment; temporal behavior under realistic system contention must be explicitly validated. Joint STER + latency verification is proposed as a candidate pre-market acceptance protocol, operationalizing FDA Draft Guidance FDA-2024-D-4488 and consistent with IEC 62304 and ISO 14971.

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